Data Flow Diagram

Date	15 March 2024
Team ID	UNM-293
Project Name	AI Comic Story Creator using Gemini Models
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., User, Gemini API)
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Generate Story, Create Images)
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Session Storage, Comic History)
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed

Data Flow Diagram for AI Comic Story Creator using Gemini Models:

Level 0 / Context Diagram:

External Entities: [User] <-> (AI Comic System) <-> [Google Gemini API (Story + Imagen)]

Data Flows: User Input (theme, characters) -> System -> Gemini API Request -> AI Response -> Comic PDF

Level 1 DFD - Detailed:

1.0 Input Handling (Process): User -> [theme, genre, chars] -> Validate Request -> Store in Session

2.0 Story Generation (Process): Session Data -> Prompt to Gemini 1.5 Flash -> Story Response -> DataStore: Story

3.0 Prompt Builder (Process): Story -> Scene Split -> Auto Image Prompts -> DataStore: Prompts

4.0 Image Generation (Process): Prompts -> Gemini Imagen API -> Comic Panel Images -> DataStore: Images

5.0 Comic Renderer (Process): Story + Images -> Layout Engine -> Final Comic View -> User + Export

6.0 Storage & Export (Data Store): Session Storage, Comic History, Output Folder -> PDF Download

DFD Diagram (Visual):

[User] --(theme, characters, genre)--> [1. Input Handling] --(validated input)--> [Session Store]

[Session Store] --(story prompt)--> [2. Gemini Story Gen] --(story)--> [Story Store]

[Story Store] --(scenes)--> [3. Prompt Builder] --(image prompts)--> [Prompt Store]

[Prompt Store] --(prompts)--> [4. Imagen API] --(panel images)--> [Image Store]

[Story Store + Image Store] --> [5. Comic Renderer] --(final comic)--> [User] + [6. Export to PDF]

External: Gemini API Service Provider <-> API Requests / Responses <-> All Processes

Signature: Dharshini.G - Team UNM-293 - AI Comic Creator with Gemini